Swing Trading Indicators

An Engineering Guide to the Math, Intuition, and Trade-Offs

Trend | SMA | EMA | Momentum | RSI | MACD | Volatility | ATR | Support/Resistance | Volume | Risk/Reward

A practical reference for understanding what each signal actually measures - and what it does not.

Prepared as a companion to a swing-trading market radar workflow

Educational material only - technical indicators are probabilistic, not predictive guarantees

Guide Map

Read this in order the first time. After that, use the indicator sections as a reference.

Part	What you will learn
1. Foundations	OHLC data, returns, timeframes, trend, signal vs noise
2. Trend filters	SMA and EMA, smoothing, lag, crossovers
3. Momentum	Rate of change, RSI, MACD and what acceleration means
4. Volatility	Return variance, standard deviation, true range and ATR
5. Market structure	Support/resistance, breakouts and volume confirmation
6. Combining signals	Scoring, confluence, dependence and avoiding indicator soup
7. Risk mathematics	Stops, targets, R multiples, expectancy and position sizing
8. Worked example	A full swing-trade setup from trend to risk decision
9. Cheat sheet	Fast reference table and common interpretation mistakes

Core mental model

Most technical indicators are deterministic transformations of price, volume, and time. They do not create new market information; they compress or re-express existing information to make certain behaviors easier to see.

1. Foundations: What the Data Actually Is

Before indicators, understand the raw time series. Every indicator in this guide is derived from a small set of observations.

1.1 OHLCV data

Field	Meaning	Typical use
Open (O)	First traded price in the period	Gap analysis, candle body
High (H)	Highest traded price	Range, resistance, true range
Low (L)	Lowest traded price	Range, support, true range
Close (C)	Last traded price	Most indicators and returns
Volume (V)	Number of shares/contracts traded	Participation and confirmation

A chart is a sampled signal

A daily chart samples the market once per trading day; a weekly chart aggregates multiple daily observations. The same asset can be bullish on one horizon and bearish on another without contradiction.

1.2 Returns: the natural unit for comparing price changes

```
Simple return:   r_t = P_t / P_(t-1) - 1

Log return:      g_t = ln(P_t / P_(t-1))
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Example: If price rises from $100 to $105, the simple return is 105/100 - 1 = 5%.

- Simple returns are intuitive for individual trades.
- Log returns are additive through time and are common in quantitative finance.
- Indicators expressed in raw dollars are not directly comparable across assets with very different prices; percentage or normalized measures often are.

1.3 Trend as a low-frequency component

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Observed price = Trend component + Cyclical movement + Noise
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- An uptrend is commonly described by higher highs and higher lows.
- A downtrend is commonly described by lower highs and lower lows.
- A range-bound market lacks a persistent directional structure.

Engineering analogy

Think of trend detection as signal processing. Moving averages behave like low-pass filters: they suppress high-frequency fluctuations so the slower-moving component becomes easier to see.

2. Moving Averages: Filtering Price Noise

Moving averages are among the simplest and most useful transformations of a price series. Their main purpose is smoothing, not prediction.

2.1 Simple Moving Average (SMA)

$$SMA_n(t) = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i}$$

- Every observation inside the lookback window has equal weight.
- Longer windows smooth more noise but react more slowly.
- Common horizons are 20, 50, 100, and 200 sessions; these are conventions, not physical constants.



Figure 1. A shorter average follows price more closely; a longer average is smoother and slower.

Condition	Typical interpretation	Caution
Price > rising SMA	Price is above a rising baseline	Can still be overextended or near resistance
Short SMA > long SMA	Faster trend exceeds slower trend	Crossovers lag and whipsaw in ranges
SMA slope > 0	Filtered trend is rising	Slope depends heavily on window length

2.2 Exponential Moving Average (EMA)

$$EMA_t = \alpha P_t + (1 - \alpha) EMA_{t-1}$$

$$\alpha = \frac{2}{n + 1}$$

- Recent observations receive more weight than older observations.
- EMA reacts faster than an equal-length SMA.
- Faster response means less lag, but also more sensitivity to noise.

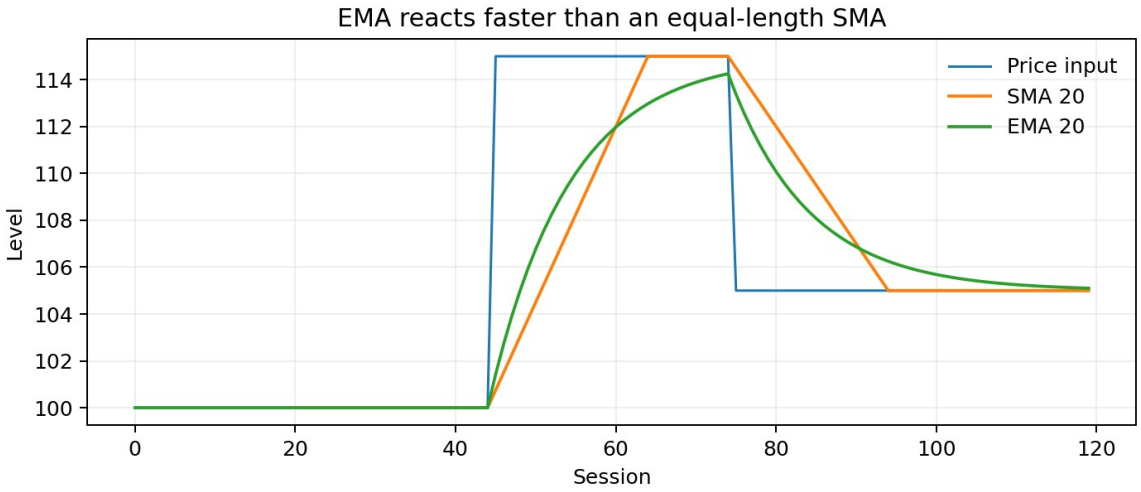


Figure 2. A step change illustrates the response-time trade-off: EMA adapts faster than SMA.

The universal trade-off

Sensitivity vs robustness: increasing sensitivity helps detect changes sooner, but also increases false signals. This is the same bias-noise trade-off seen in many estimation problems.

3. Momentum: Direction Is Not Enough

Trend asks where price is going. Momentum asks how strongly or how quickly it is moving relative to its recent history.

3.1 Raw momentum and Rate of Change (ROC)

$$\text{Momentum}_n(t) = P_t - P_{(t-n)}$$

$$\text{ROC}_n(t) = P_t / P_{(t-n)} - 1$$

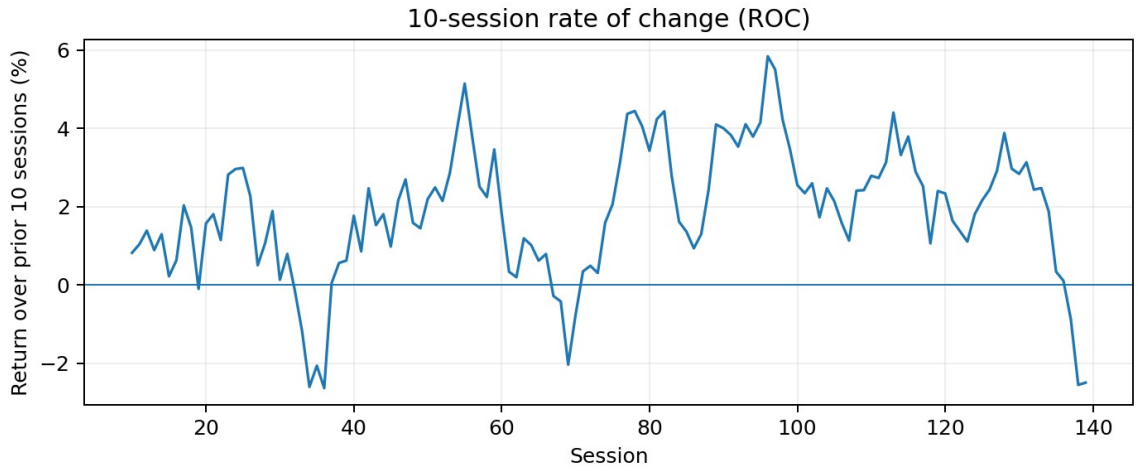


Figure 3. ROC above zero means price is above its level n sessions ago; the magnitude measures the size of that move.

- Positive momentum: current price is above its comparison point.
- Rising positive momentum: the move is strengthening.
- Falling positive momentum: price may still be rising, but the rate of increase is weakening.

3.2 RSI: Relative Strength Index

RSI is a bounded momentum oscillator introduced by J. Welles Wilder. It compares the magnitude of recent gains with recent losses.

$$\text{RS} = \text{Average Gain} / \text{Average Loss}$$

$$\text{RSI} = 100 - 100 / (1 + \text{RS})$$

$$0 <= \text{RSI} <= 100$$

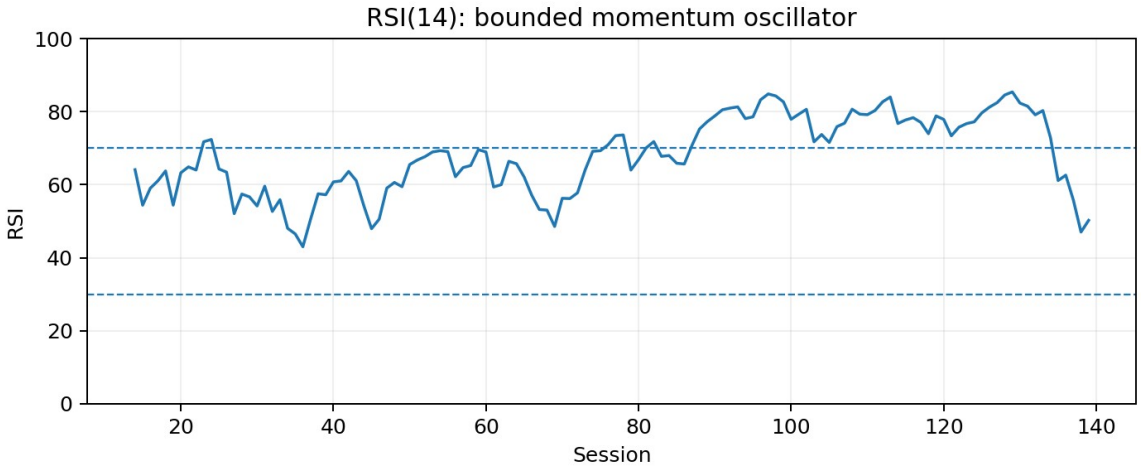


Figure 4. RSI is bounded; 70 and 30 are common reference levels, not guaranteed reversal points.

RSI zone	What it often means	What it does NOT mean
> 70	Recent upside momentum has been strong	Price must immediately fall
50-70	Positive momentum dominates	A buy signal by itself
30-50	Negative momentum dominates	A sell signal by itself
< 30	Recent downside momentum has been strong	Price must immediately rebound

Why “overbought” is misleading

In a strong trend, RSI can remain above 70 for a long time. A high RSI is evidence of strong recent buying pressure, not a timer counting down to a crash.

3.3 MACD: Comparing Two Trend Speeds

$$MACD_t = EMA_{12}(t) - EMA_{26}(t)$$

$$Signal_t = EMA_9(MACD_t)$$

$$Histogram_t = MACD_t - Signal_t$$

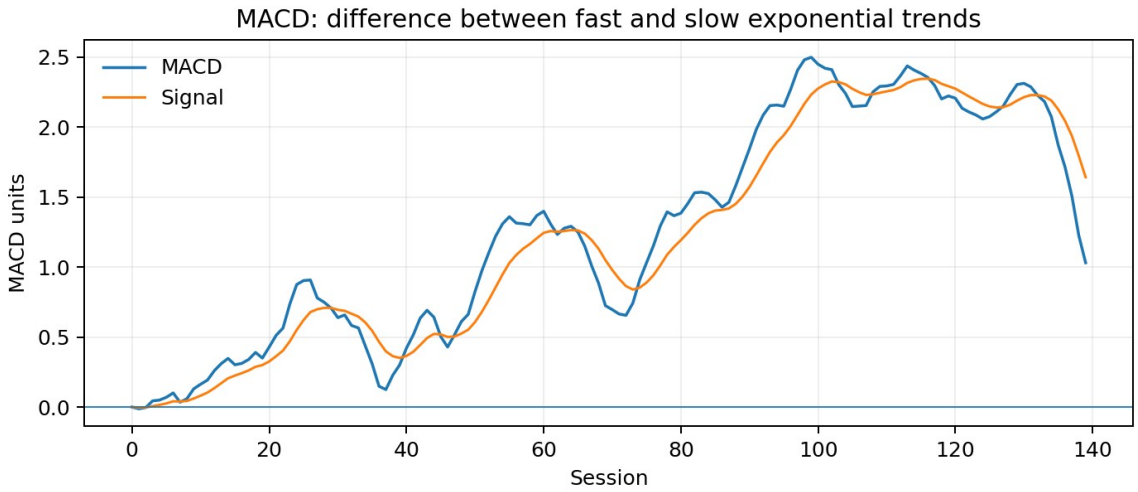


Figure 5. MACD compares a faster exponential trend with a slower one. The signal line smooths the MACD itself.

Observation	Interpretation	Caution
MACD > 0	Fast EMA is above slow EMA	Does not guarantee price is “cheap” or early
MACD crosses above signal	Momentum is improving relative to its own recent baseline	Crosses can whipsaw in sideways markets
Histogram increases	Distance between MACD and signal is expanding	Acceleration can fade quickly
Bearish divergence	Price makes higher high while momentum does not	Divergence may persist without immediate reversal

A useful interpretation

MACD is a difference of filtered signals. Instead of thinking “buy/sell crossover,” ask: is the short-horizon filtered trend strengthening or weakening relative to the longer-horizon trend?

4. Volatility: How Much Price Moves

Volatility is about magnitude, not direction. A highly volatile asset can rise or fall violently; a low-volatility asset can drift steadily.

4.1 Statistical volatility

$$\text{Sample variance: } s^2 = \frac{1}{n-1} \sum (r_i - \bar{r})^2$$

$$\text{Sample volatility: } s = \sqrt{s^2}$$

$$\text{Annualized daily volatility} \approx s_{daily} \times \sqrt{252}$$

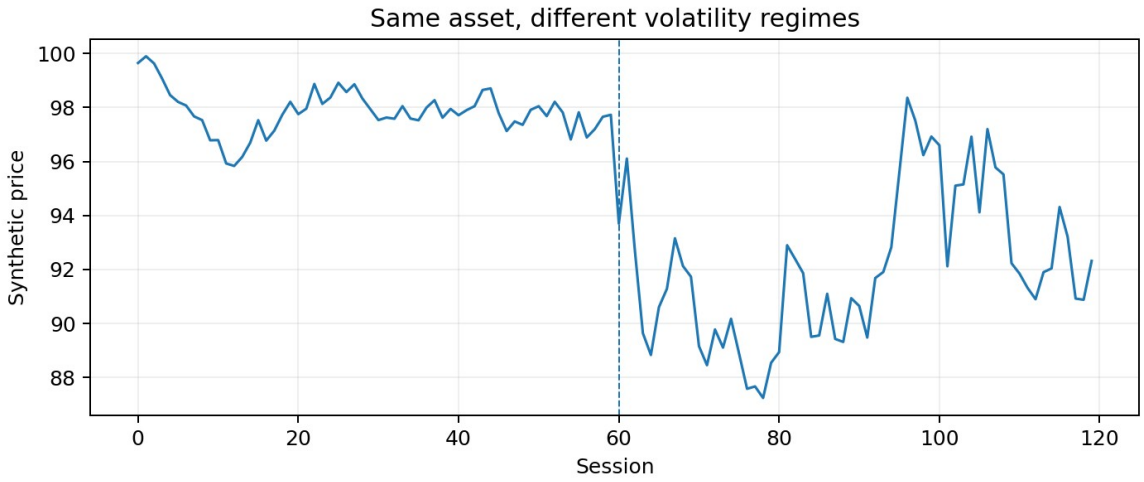


Figure 6. Volatility can change through time; using one constant risk estimate can therefore be misleading.

- Volatility rises when returns become more dispersed around their mean.
- Annualization uses a square-root-of-time approximation and assumes limited serial dependence.
- Historical volatility is backward-looking; future volatility can differ substantially.

4.2 True Range and ATR

ATR converts recent price movement into an absolute price-unit estimate. It is especially useful for stop placement and position sizing.

$$TR_t = \max(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|)$$

$$ATR_n = \text{smoothed average of True Range over } n \text{ periods}$$

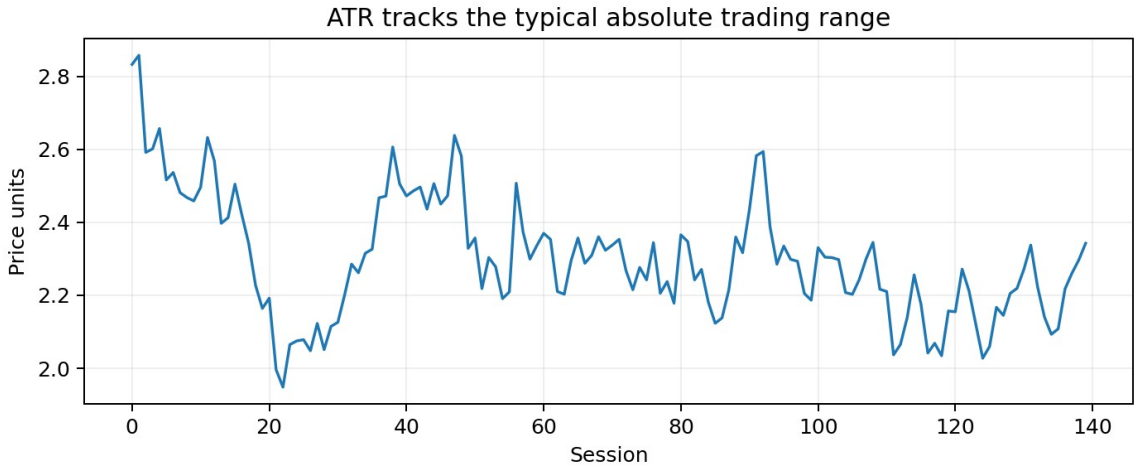


Figure 7. ATR rises when the typical trading range expands and falls when price movement contracts.

Example	Value
Current price	$100
ATR(14)	$4
Stop distance of $1	0.25 ATR - probably inside normal noise
Stop distance of $6	1.5 ATR - more room, but more capital at risk

$$\text{Volatility-adjusted stop example: Stop} = \text{Entry} - k * \text{ATR}$$

$$\text{Position size} = \frac{\text{Maximum dollar risk}}{\text{Stop distance}}$$

> **ATR is not directional**
> ATR = $4 does not mean the asset will rise $4. It means recent trading ranges have been roughly that size. Direction must come from other information.

5. Support, Resistance, and Market Structure

Support and resistance are not deterministic barriers. They are price regions where historical trading behavior suggests supply or demand may become important.

5.1 A practical mathematical view

- Detect local extrema: points where price is high or low relative to neighboring observations.
- Cluster nearby extrema into price zones instead of treating each level as exact.
- Count touches, recency, rejection magnitude, and volume as possible strength features.
- Treat a breakout as a regime change hypothesis, not proof.

Illustrative zone score = w1*touches + w2*recency + w3*rejection_strength + w4*volume_confirmation

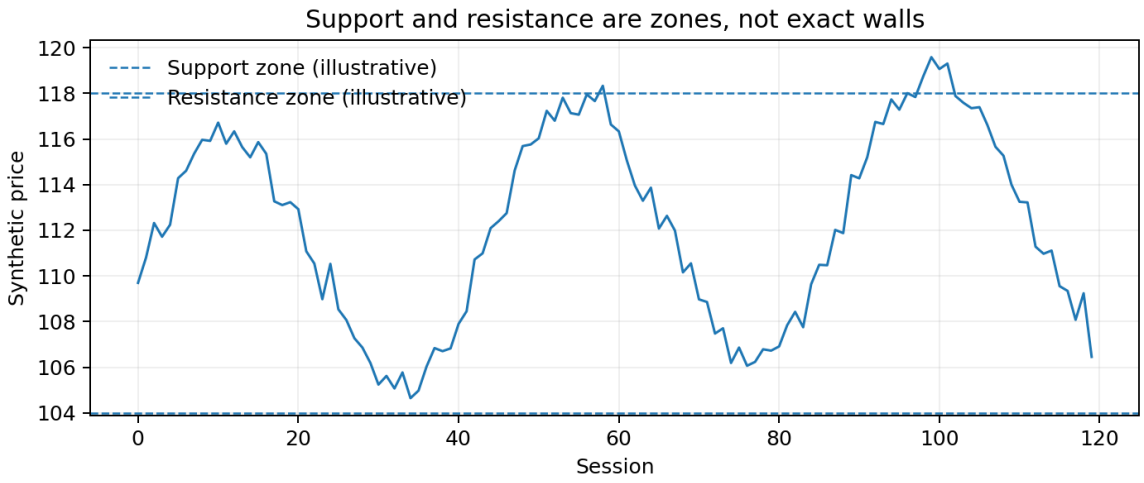


Figure 8. Real support/resistance is better modeled as a zone with uncertainty than a single exact price.

5.2 Breakouts

Breakout condition (simple): P_t > recent resistance

- A breakout can fail and return into the prior range (“false breakout”).
- A closing-price filter can reduce intraday noise but delays confirmation.
- Volume or range expansion can provide additional context, but neither guarantees continuation.

6. Volume: Participation Behind the Move

Volume measures trading activity. It is often used as a confirmation variable because the same price move can occur with very different levels of participation.

$$\text{Relative Volume (RVOL)} = \frac{\text{Current Volume}}{\text{Average Volume over lookback window}}$$

Scenario	Possible interpretation
Price breakout + RVOL >> 1	Broad participation may support the breakout hypothesis
Price breakout + low volume	Move may be less convincing or simply occur in thin trading
Price falling + high volume	Strong selling participation or forced de-risking may be present
Price flat + high volume	Potential absorption: a lot of trading without much price progress

Volume is context, not a vote count

Every completed trade has both a buyer and a seller. “High buying volume” is shorthand for aggressive buying pressure pushing transactions to higher prices, not that buyers numerically outnumber sellers.

7. Multiple Timeframes: Daily vs Weekly

Trend is scale-dependent. A daily pullback can occur inside a strong weekly uptrend. This is common and often useful for swing-trading setups.

Weekly trend	Daily trend	Typical reading
Up	Up	Trend aligned; continuation setup may be mature or extended
Up	Down	Possible pullback inside larger uptrend
Down	Up	Counter-trend bounce inside larger downtrend
Down	Down	Trend aligned to downside

The question to ask

Do not ask “Is the stock bullish?” Ask “Bullish on what time horizon?” Technical labels are incomplete without a sampling interval.

8. Combining Indicators: Scoring and Confluence

A market radar often compresses several measurements into a classification. Conceptually, this is feature engineering followed by a rule or score.

$$\text{Feature vector } x = [\text{trend}, \text{momentum}, \text{volatility}, \text{structure}, \text{volume}, \ldots]$$

$$\text{Score} = w_1 x_1 + w_2 x_2 + \ldots + w_k x_k$$

$$\text{Classification} = \text{threshold}(\text{Score})$$

- A score is a summary, not an oracle.
- Thresholds trade off sensitivity and false positives.

- Weights and thresholds should ideally be validated out of sample, not tuned until historical results look perfect.
- Regime changes can invalidate relationships that worked historically.

8.1 Confluence is useful - but indicators are correlated

Many indicators are transformations of the same price series. Ten bullish indicators do not represent ten independent pieces of evidence.

Indicator	Primary input	Main information emphasized
SMA / EMA	Price	Smoothed trend
ROC	Price	Change over a fixed horizon
RSI	Price changes	Balance of recent gains vs losses
MACD	Price	Difference between fast and slow filtered trends
Volatility	Returns	Dispersion of price changes
ATR	OHLC	Typical absolute range
Support/resistance	Price extrema	Repeated market structure zones
Relative volume	Volume	Participation relative to baseline

Avoid indicator soup

If SMA, EMA, MACD, and RSI all look bullish, much of that agreement may simply reflect the same recent price rise. Prefer a small set of indicators that answer different questions: trend, momentum, volatility, structure, and participation.

8.2 Common statistical traps

- Look-ahead bias: using information that would not have been available at the decision time.
- Data snooping: trying many indicators/settings and reporting only the combination that worked.
- Overfitting: fitting noise in one market regime rather than a stable pattern.
- Selection bias: analyzing only assets that survived or became successful.
- Ignoring transaction costs, slippage, taxes, and liquidity constraints.
- Treating correlated indicators as independent confirmation.

9. Risk/Reward, Expectancy, and Position Sizing

A trader does not need to predict correctly most of the time if the payoff distribution is favorable and losses are controlled.

9.1 R multiple

```
Risk per share = Entry - Stop      (for a long trade)

Potential reward = Target - Entry

Reward/Risk ratio = (Target - Entry) / (Entry - Stop)
```

Example input	Value
Entry	$100
Stop	$95

Example input	Value
Target	$115
Risk	$5 per share
Potential reward	$15 per share
Reward/Risk	3.0

9.2 Expectancy

```
Expectancy = P(win)*AverageWin - P(loss)*AverageLoss
```

```
Example = 0.40*(+3R) + 0.60*(-1R) = +0.60R per trade
```

Why this matters

A system can lose more often than it wins and still have positive expectancy. Conversely, a high win rate can be unprofitable if occasional losses are much larger than wins.

9.3 Position sizing

```
Dollar risk budget = Account equity * Risk fraction
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Shares = Dollar risk budget / (Entry - Stop)
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Example: A $20,000 account risks 0.5% ($100). Entry is $100 and stop is $96, so risk per share is $4. Position size = $100 / $4 = 25 shares.

- Wider stops require smaller positions for the same dollar risk.
- Volatility-based stops can make risk more comparable across assets.
- The stop level should reflect where the trade thesis is invalidated, not only what loss feels comfortable.

10. Worked Example: From Radar Signal to Trade Decision

This example is illustrative and uses hypothetical data. The goal is to show how the indicators fit together, not to recommend a security.

Step	Observation	Interpretation
1. Weekly trend	Price > rising 50-week average	Larger horizon is constructive
2. Daily trend	Price pulled back below 20-day EMA but remains above 50-day SMA	Short-term weakness inside medium-term uptrend
3. RSI	RSI fell from 72 to 44	Momentum cooled; no longer extremely stretched
4. MACD	Histogram stops falling and turns upward	Downside momentum is decelerating
5. Structure	Price is near a prior breakout/support zone	A logical area for the thesis to be tested
6. ATR	ATR = $3.20	Normal daily movement is large enough to require room
7. Volume	Rebound day has RVOL = 1.6	Participation is above recent baseline

```
Hypothetical entry = $120
```

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Hypothetical stop = $114   -> risk = $6/share ~= 1.9 ATR

Hypothetical target = $138 -> reward = $18/share

Reward/Risk = 18 / 6 = 3.0
```

Decision logic

The setup is not “good” because RSI or MACD says so. It is coherent because different categories align: larger trend, pullback structure, recovering momentum, reasonable volatility-adjusted invalidation, and acceptable payoff asymmetry.

10.1 Pre-trade checklist

1. What is the dominant trend on the weekly and daily horizons?
2. Is price near a meaningful structure level, or am I chasing after a large move?
3. What does momentum say: strengthening, weakening, or neutral?
4. What is the current volatility regime and ATR?
5. Where is the trade thesis invalidated?
6. What is the reward/risk ratio to a realistic target?
7. How many shares fit the predefined dollar-risk budget?
8. Are volume, earnings, or major events likely to change the risk profile?
9. Am I counting several correlated indicators as if they were independent evidence?

11. Indicator Cheat Sheet

Indicator	Core formula / idea	Answers	Main failure mode
SMA	Arithmetic mean of last n prices	What is the smoothed trend?	Lag; whipsaw in ranges
EMA	Recursive weighted average	What is the faster smoothed trend?	Noise sensitivity
ROC	P_t / P_(t-n) - 1	How much has price changed?	Lookback dependence
RSI	100 - 100/(1+AvgGain/AvgLoss)	Who dominated recent moves?	Overbought/oversold misread as reversal
MACD	EMA12 - EMA26	Is fast trend stronger than slow trend?	Lag and crossover whipsaw
Std. vol	Std. dev. of returns	How dispersed are returns?	Backward-looking, regime shifts
ATR	Smoothed True Range	What is the typical absolute range?	Not directional
Support/Resistance	Clusters of repeated extrema	Where may supply/demand matter?	Subjective zones; false breakouts
RVOL	Volume / average volume	Is participation unusually high?	Event-driven spikes, venue effects
R:R	Potential reward / planned risk	Is payoff asymmetric enough?	Target may be unrealistic
Expectancy	pW*avgW - pL*avgL	Is the process profitable on average?	Requires stable, honest estimates

11.1 The five-question framework

Question	Best indicator family
Direction: where is price trending?	SMA / EMA / market structure
Momentum: how strong is the move?	ROC / RSI / MACD

Question	Best indicator family
Volatility: how large is normal noise?	Return volatility / ATR
Structure: where can the thesis fail or break out?	Support / resistance / prior highs and lows
Participation: how much trading activity supports the move?	Volume / relative volume

If you remember only one thing

Indicators are measurement tools. A disciplined process asks a small number of distinct questions, quantifies risk before entry, and evaluates results over many trades. It does not treat a green signal as certainty.

12. Compact Glossary

Term	Meaning
Bullish	Evidence or expectation consistent with rising prices.
Bearish	Evidence or expectation consistent with falling prices.
Pullback	A temporary move against the dominant trend.
Breakout	Price moves beyond a prior range or resistance/support region.
Whipsaw	A signal reverses quickly, often causing repeated small losses.
Divergence	Price and an indicator move in different directions.
Lag	Delay between a market change and an indicator response.
Regime	A period with different statistical behavior, such as high vs low volatility.
R multiple	Profit/loss normalized by the amount initially risked.
Confluence	Several different categories of evidence support the same thesis.

13. Further Reading

Good sources if you want to move from practical use to deeper technical study:

- J. Welles Wilder Jr., New Concepts in Technical Trading Systems (1978) - original RSI and ATR framework.
- John J. Murphy, Technical Analysis of the Financial Markets - broad reference on trends, chart structure, volume, and indicators.
- Gerald Appel, Technical Analysis: Power Tools for Active Investors - MACD and trend/momentum applications.
- Ernest P. Chan, Algorithmic Trading - a quantitative bridge from indicators to systematic testing and statistical validation.
- Robert Carver, Systematic Trading - risk sizing, diversification, and robust rule-based decision making.

Educational disclaimer: This guide explains market-analysis concepts and does not constitute investment, tax, or financial advice. Historical relationships and indicator behavior do not guarantee future results.